

program4.py

```
1  from threading import Thread, current_thread
2  from multiprocessing import Process, Pipe, Value
3
4  processes = [
5      {"pid": "P1", "arrival": 0, "burst": 7},
6      {"pid": "P2", "arrival": 2, "burst": 4},
7      {"pid": "P3", "arrival": 4, "burst": 1},
8      {"pid": "P4", "arrival": 5, "burst": 4},
9  ]
10
11 intervals = [
12     ("P1",0,2), ("P2",2,4), ("P1",4,6),
13     ("P3",6,7), ("P2",7,9), ("P4",9,11),
14     ("P1",11,13), ("P4",13,15), ("P1",15,16)
15 ]
16
17 def calculate_metrics():
18     print("\n--- SCHEDULING METRICS ---")
19     print("PID AT BT CT TAT WT RT")
20     tat_total = wt_total = rt_total = 0
21     for p in processes:
22         runs = [x for x in intervals if x[0] == p["pid"]]
23         first_start = runs[0][1]
24         completion = runs[-1][2]
25         tat = completion - p["arrival"]
26         wt = tat - p["burst"]
27         rt = first_start - p["arrival"]
28         tat_total += tat
29         wt_total += wt
```

```
• yashasvi_bhagour@LAPTOP-CJHR9PD4:~/OS-Fourth-Program$ pwd
/home/yashasvi_bhagour/OS-Fourth-Program
• yashasvi_bhagour@LAPTOP-CJHR9PD4:~/OS-Fourth-Program$ python3 program4.py
```

--- SCHEDULING METRICS ---

PID	AT	BT	CT	TAT	WT	RT
P1	0	7	16	16	9	0
P2	2	4	9	7	3	0
P3	4	1	7	3	2	2
P4	5	4	15	10	6	4

Average TAT = 9.0

Average WT = 5.0

Average RT = 1.5

--- ROUND ROBIN GANTT CHART ---

P1	P2	P1	P3	P2	P4	P1	P4	P1	
0	2	4	6	7	9	11	13	15	16

--- THREAD DEMO ---

Thread 1 running | Thread ID: 127258280851136

Thread 2 running | Thread ID: 127258272458432

Both threads completed.

--- PIPE IPC DEMO ---

Parent received: Hello Parent - message from Child

--- SHARED MEMORY DEMO ---

Before child process: 5

After child process : 15

```
• yashasvi_bhagour@LAPTOP-CJHR9PD4:~/OS-Fourth-Program$
```